

MECH321 – SOLID MECHANICS II

Course Syllabus – Fall 2026

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This is your course syllabus. Please download the file and keep it for future reference.

TEACHING TEAM

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory. See: <http://www.queensu.ca/encyclopedia/t/traditional-territories>

INCLUSIVITY STATEMENT

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: <https://www.queensu.ca/vpcei/support/overview>.

COURSE INSTRUCTOR

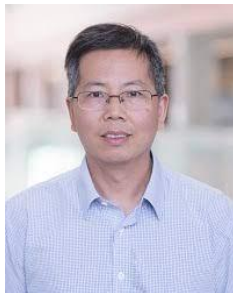
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Office hours: TBD



MECH321 (F 3-0-0.5 3.5)

COURSE DESCRIPTION

This course continues the study of solid mechanics that was introduced in second year. By the end of this course, learners should be able to: 1) Calculate the total normal and shear stress at a point and sketch the stress distributions on a cross-section of a structural component (such as a crank) experiencing 3D combined (axial, transverse and/or moment causing) loads and non-symmetric loads; 2) Calculate the residual normal or shear stress at a point and sketch the stress distribution on a cross-section of a structural component that is experiencing axial, torsional and/or bending loads followed by unloading; 3) Calculate the normal or shear stress at a point on a cross-section of a structural component that is under load (axial, torsional and/or bending) and is supported in a statically indeterminate configuration (using force balance equations together with compatibility equations derived from known boundary conditions); 4) Calculate the normal or shear stress at a point on a cross-section of a structural component that is under load (axial, torsional and/or bending) and contains one or more locations of stress concentration; 5) Calculate, using general equations and/or graphically using a Mohr's circle, the normal and shear stress and/or strain transformations at a point within a structural component under load as a function of the orientation relative to a fixed coordinate system and find the maximum in-plane normal and shear stress and/or strain; 6) Calculate the deflections and angles of deflection at any point on a transversely loaded beam of uniform cross-section using the principle of superposition and the standard equations for single loads acting on simply supported beams 7) Solve for critical loads in terms of buckling for concentrically and eccentrically loaded columns; 8) Calculate the optimum dimensions (design) for shafts and beams under combined 3D loading based on specified material failure criteria; 9) Design mechanism or structural components to withstand all forces for given loads, maximum deflection tolerances, factor of safety and material properties; 10) Calculate the deflections and angles of deflection at any point on a beam or truss using Energy Methods.

(0/0/0/42/0) (Mathematics/Natural Sciences/Complementary Studies/Engineering Science/Engineering Design)

MODE OF INSTRUCTION

The delivery mode of this course is in-person.

COURSE LOCATION AND TIME

LECTURES: MON. 2:30-3:30pm, TUE. 4:30-5:30pm THU. 3:30-4:30 pm

TUTORIALS: FRI. 4:30-5:30pm.

LOCATION: DUNNING HALL AUD.

PRE-REQUISITE KNOWLEDGE

This course is designed for learners who already possess a full knowledge of statics (the force analysis of static structures) and an introductory knowledge of solid mechanics.

Prerequisite courses: CIVL 220 or MECH 221

COURSE LEARNING OUTCOMES (CLO)

By the end of this course, students should be able to:

CLO	DESCRIPTION	INDICATOR
CLO 1	Calculate the total normal and shear stress at a point and sketch the stress distributions on a cross-section of a structural component (such as a crank) experiencing 3D combined (axial, transverse and/or moment causing) loads and non-symmetric loads.	KB-ES (Solid Mechanics)
CLO 2	Calculate the residual normal or shear stress at a point and sketch the stress distribution on a cross-section of a structural component that is experiencing axial, torsional and/or bending loads followed by unloading.	KB-ES (Solid Mechanics)
CLO 3	Calculate the normal or shear stress at a point on a cross-section of a structural component that is under load (axial, torsional and/or bending) and is supported in a statically indeterminate configuration (using force balance equations together with compatibility equations derived from known boundary conditions).	KB-ES (Solid Mechanics)
CLO 4	Calculate the normal or shear stress at a point on a cross-section of a structural component that is under load (axial, torsional and/or bending) and contains one or more locations of stress concentration.	KB-ES (Solid Mechanics)
CLO 5	Calculate, using general equations and/or graphically using a Mohr's circle, the normal and shear stress and/or strain transformations at a point within a structural component under load as a function of the orientation relative to a fixed coordinate system and find the maximum in-plane normal and shear stress and/or strain.	KB-ES (Solid Mechanics)
CLO 6	Calculate the deflections and angles of deflection at any point on a transversely loaded beam of uniform cross-section using the principle of superposition and the standard equations for single loads acting on simply supported beams.	KB-ES (Solid Mechanics)
CLO 7	Solve for critical loads in terms of buckling for concentrically and eccentrically loaded columns.	KB-ES (Solid Mechanics)

CLO 8	Calculate the optimum dimensions (design) for shafts and beams under combined 3D loading based on specified material failure criteria	DE-Solutions
CLO 9	Analysis and design mechanism or structural components to withstand all forces for given loads, maximum deflection tolerances, factor of safety and material properties.	DE-Strategies DE-Assess DE-Solutions
CLO 10	Calculate the deflections and angles of deflection at any point on a beam or truss with Energy Methods.	KB-ES (Solid Mechanics)
CLO 11	Present engineering design or analysis results efficiently in written documents.	CO-Written CO-Graphics CO-Documentation

COURSE EVALUATION

Assessment Weighting

Assessment Tool	Due Date (Eastern Time)	Weight	Alignment with CLOs
Midterm 1	Week 5	20%	1, 2, 3, 4
Midterm 2	Week 9	20%	1, 2, 3, 5
Final Exam	TBD	40%	1-10
Design Project		20%	3,4,5,6,8,9,10,11
Report 1	Week 2	4%	3,4,9,11
Report 2	Week 8	6%	3,4,5,6,8,9,11
Report 3	Week 12	10%	3,4,5,6,8,9,10,11
		100%	

ASSESSMENT DESCRIPTIONS

Midterm Test:

There are TWO midterm tests (closed book) in this course. These tests are designed to provide learners with immediate feedback on their knowledge. Students must write their exam on the day and time scheduled. **You should not schedule vacations, travel, etc. during the test.** Detailed policies regarding midterm tests will be provided in OnQ.

Project Reports:

There are THREE phase reports for the project in this course. The first and second reports will require you to analyze the given structure under specified load. The third report will require you to modify the structure, and validate your modification. More details about these reports can be found in onQ.

Final Exam

The final exam is closed book. Students must write their exam on the day and time scheduled by the University. You should not schedule vacations, travel, etc. during the exam period. The [Term and Session Dates](#) will indicate the final exam period session dates in each term.

GRADING

(STANDARD TEXT): All assessments in this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established [Grade Point Index](#).

Feedback on Assessments

The teaching team will provide feedback on graded activities. You can expect feedback on your assessments within seven days of the due date.

Accessing Your Final Grade

Your final grades will show on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

COURSE MATERIALS

Required Textbook

- Mechanics of Materials, 10th SI Edition. By: R.C. Hibbeler. ISBN-13: 978-1-292-17820-2

Other editions can be used, but tutorial problems are taken from the 10th edition.

Suggested Time Commitment

This course represents a study period of one term spanning 12 weeks. Learners can expect to invest on average 7-9 hours per week in this course. Learners who adhere to a predetermined study schedule are more likely to successfully complete the course.

WEEKLY COURSE LEARNING OUTCOMES

Week	Learning Outcomes	Assessment
1	General Introduction & Review [CLO1] [CLO6] By the end of this week, learners will be able to: - Determine normal stress and strain under axial loads [CLO1] - Determine shear stresses and strain under torsional load [CLO1] - Determine normal and shear stresses under bending and transverse loads [CLO1] [CLO6]	
2	Combined Load/Axial Load [CLO3] [CLO6] By the end of this week, learners will be able to: - Determine supports and normal stress for statically indeterminate members under axial loading [CLO3] - Determine deflection and angles of deflection at any point on a transversely loaded beam using superposition principle [CLO6]	
3	Residual stress from Axial load [CLO4] [CLO2] [CLO3] By the end of this week, learners will be able to: - Determine support loads and normal stress for statically indeterminate members under axial load and /or thermal stress [CLO3] - Determine normal stress affected stress concentration features [CLO4] - Determine inelastic deformation of members under axial loads [CLO2] - Determine residual stress/strain for axially loaded members [CLO2]	Phase 1 Report [CLO3] [CLO4] [CLO9] [CLO11]

Week	Learning Outcomes	Assessment
4	Torsional loads [CLO1] [CLO3] By the end of this week, learners will be able to: - Determine support loads and shear stresses for statically indeterminate shafts [CLO3] - Determine shear stress under torsion for circular shafts, thin-walled tubes [CLO1]	
5	Torsional and Bending loads [CLO1] [CLO2] By the end of this week, learners will be able to: - Determine shear and moment diagram using graphic method [CLO1] - Determine normal stress and transverse shear stress [CLO1] - Determine shear stresses and strain due to inelastic deform under torsional load [CLO1] [CLO2] - Determine residual stress/strain for torsionally loaded members [CLO2]	Midterm Test 1: [CLO1] [CLO2] [CLO3] [CLO4] [CLO6]
6	Inelastic bending & stress/strain transformation [CLO1] [CLO2] By the end of this week, learners will be able to: - Determine normal stress and strain under unsymmetrical bending loads [CLO1] - Determine normal stress and strain for curved beams under bending loads [CLO1] - Determine residual stress in bending [CLO1] [CLO2]	

Week	Learning Outcomes	Assessment
7	Strain transformation [CLO5] By the end of this week, learners will be able to: • Conduct strain transformation [CLO5] • Determine strain in desired orientation from strain rosette [CLO5] • Review stress transformation [CLO5]	
8	Failure theories [CLO1] [CLO5] By the end of this week, learners will be able to: • Use failure theories to determine the safety level for ductile members under axial, torsional and bending loads [CLO1] [CLO5] • Determine strains / stresses from strain Rosettes readings [CLO1]	Phase 2 Report [CLO3] [CLO4] [CLO5] [CLO6] [CLO8] [CLO9] [CLO11]
9	Failure theories, beam design[CLO1][CLO5][CLO6][CLO8][CLO9] By the end of this week, learners will be able to: • Use failure theories to determine the safety level for brittle members under axial, torsional and bending loads [CLO1] [CLO5] • Design fully stressed beam [CLO8] [CLO9] • Determine slope and deflection of the beam or shaft under multiple loads[CLO6] • Design and optimize beam satisfying safety requirement [CLO1] [CLO5] [CLO8]	Midterm Test 2: [CLO1] [CLO2] [CLO3] [CLO5]

Week	Learning Outcomes	Assessment
10	Shaft design and Columns [CLO1][CLO5][CLO7][CLO8][CLO9][CLO10] By the end of this week, learners will be able to: - Determine shaft diameter using specific failure theory [CLO1][CLO5][CLO8][CLO9] - Identify the critical load / stress of columns [CLO7] - Determine slope and deflection for a structure or beam/shaft using superposition[CL9][CLO10]	
11	Conservation of Energy [CLO1][CLO9][CLO10] By the end of this week, learners will be able to: - Determine slope and deflection for a structure or beam/shaft using energy conservation principle[CL9][CLO10] - Determine the maximum load/stress of members caused from impact loadings[CLO1][CLO10]	
12	Virtual Work [CLO10] By the end of this week, learners will be able to: Determine deflection/slope of structure or beam/shaft using energy method [CLO10]	Phase 3 report [All CLOs]

COURSE COMMUNICATION

QUESTIONS ABOUT COURSE MATERIAL

Questions or comments regarding the course material that can be of benefit to other students should be posted in the Q&A forum on the class website. The instructor, TAs, and students are encouraged to answer these questions directly in the discussion forum for the benefit of everyone in the course.

COURSE ANNOUNCEMENTS

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ. Please sign up to be automatically notified by email when the

instructor posts new information in the Announcements section. Instructions on how to modify your notifications are found in the **Begin Here** section of the onQ course site.

OFFICE HOURS

In addition to interaction in the Q&A discussion forums, you will have the opportunity to interact with either a TA or the instructor during office hours. The instructor will provide a schedule of availability at the beginning of the term.

CONFIDENTIAL MATTERS

If you have a confidential matter you would like to discuss with your instructor, their contact details are on the first page of this document. Expect email replies within 48 hours.

ABSENCES (ACADEMIC CONSIDERATIONS) AND MISSED ASSIGNMENTS

Students who miss assignments, quizzes, classes, labs, tutorials or other course deliverables due to unforeseen personal circumstances outside of their control can apply for Academic Consideration. For information on academic considerations due to extenuating circumstances, please review the information on the [Smith Engineering website](#). Note that unacceptable reasons include extracurricular activities, work, travel plans, scheduling conflicts, generally being behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as **travel and work are not acceptable reasons for granting academic considerations.**

students should send academic considerations to lai@queensu.ca , and policies or standards around academic consideration options (eg. reweighting of grades or alternative assessment options, in the event of an approved academic consideration)

LATE SUBMISSION POLICY

In the event of extenuating circumstances, you must follow the policies for requesting an academic consideration (as described above) or engage the appropriate Academic Accommodation process (see Academic Accommodation section below). In the absence of an approved consideration request, or accommodation in VENTUS, normal late penalty will apply as described in the assignment descriptions or any course/departamental policies.

REQUESTING ACADEMIC CONSIDERATION RELATED TO GROUP WORK

Group submissions may pose challenges when one or more members of the group experience extenuating circumstances.

Please have one member of your group reach out to your instructor if a group member is experiencing an extenuating circumstance that may impact your groups ability to submit a deliverable on time.

STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES

NETIQUETTE

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

The following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.
5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

STUDENT CODE OF CONDUCT

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation. All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.

<https://www.queensu.ca/nonacademicmisconduct/policies>

COPYRIGHT

Course materials created by the course instructor, including all slides, presentations, synchronous and asynchronous course recordings, handouts, tests, exams, and other similar course materials, are the intellectual property of the instructor. It is a departure from academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the instructor's **express**

consent. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights and, with respect to recordings, potentially privacy violations of other students.

ACADEMIC INTEGRITY

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

- Act at all times with devotion to the high ideals of personal honour and professional integrity.
- Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the [Professional Engineers Ontario Code of Ethics](#). Information on policies concerning academic integrity is available in the [Queen's University Code of Conduct](#), in the [Senate Academic Integrity Policy Statement](#), on the [Smith Engineering website](#), and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

PLAGIARISM/USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GENAI) TOOLS:

Students must submit their own work and cite the work that is not theirs. Any other use constitutes a Departure from Academic Integrity.

GenAI tools are NOT permitted in this course: Use of GenAI tools are not allowed in any part of student work for this course. Submitting AI-generated content constitutes a Departure from Academic Integrity as defined by University Academic Integrity procedures.

TURNITIN STATEMENT

This course makes use of Turnitin, a third-party application that helps maintain standards of excellence in academic integrity. Normally, students will be required to submit their course assignments through onQ to Turnitin. In doing so, students' work will be included as source documents in the Turnitin reference database, where they will be used solely for the purpose of detecting plagiarized text in this course. Data from submissions is also collected and analyzed by Turnitin for detecting Artificial Intelligence (AI)-generated text. These results are not reported to your instructor at this time but could be in the future.

Turnitin is a suite of tools that provide instructors with information about the authenticity of submitted work and facilitates the process of grading. The similarity report generated after an assignment file is submitted produces a similarity score for each assignment. A similarity score is the percentage of writing that is similar to content found on the internet or the Turnitin extensive database of content. Turnitin does not determine if an instance of plagiarism has occurred. Instead, it gives instructors the information they need to determine the authenticity of work as a part of a larger process.

Please read Turnitin's [Privacy Policy](#), [Acceptable Use Policy](#) and [End-User License Agreement](#), which govern users' relationship with Turnitin. Also, please note that Turnitin uses cookies and other tracking technologies; however, in its service contract with Queen's Turnitin has agreed that neither Turnitin nor its third-party partners will use data collected through cookies or other tracking technologies for marketing or advertising purposes.

For further information about how you can exercise control over cookies, see [Turnitin's Privacy Policy](#).

Turnitin may provide other services that are not connected to the purpose for which Queen's University has engaged Turnitin. Your independent use of Turnitin's other services is subject solely to Turnitin's Terms of Service and Privacy Policy, and Queen's University has no liability for any independent interaction you choose to have with Turnitin.

Portions of this document have been adapted, with permission, from the University of Toronto Centre for Teaching Support and Innovation tip sheet "[Turnitin: An Electronic Resource to Deter Plagiarism](#)".

INVALID EXAMS

An exam may be declared invalid in case of an interruption in an in-person examination; if the instructions in a remote or online exam were not followed; if the student uploads wrong materials; or if a situation arises where the integrity of the exam cannot be verified. If an exam is declared invalid, the student may be granted a re-write.

ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you, including the [Library](#), [Student Academic Success Services \(Learning Strategies and Writing Centre\)](#), and [Career Services](#). Learners are encouraged to visit the Smith Engineering [Current Students](#) web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

ABSENCES - ACADEMIC CONSIDERATIONS

Academic Consideration is a process by which students request validated time away from their academics when they are experiencing an extenuating circumstance or personal circumstances beyond a student's control that interfere with their ability to complete academic requirements.

For additional information and to apply for academic consideration, please review the information on the [Smith Engineering's Academic Consideration \(Absences\) website](#).

ACADEMIC ACCOMMODATIONS

Academic Accommodations are available when students need an adjustment to their learning environment due to a reason related to a Human Rights protected ground. Below there is information on the most common Academic Accommodations requested by students. For more information, please see [Smith Engineering's Academic Accommodation website](#).

ABSENCES – REQUESTS FOR EXCUSED ABSENCES FOR SIGNIFICANT EVENTS (REASE)

Students participating in a Varsity Athletics event, the student reserve forces, a non-varsity event at the provincial, national, or international level, or an event to which you were invited as a distinguished guest can submit a Request for Excused Absence for Significant Event following the REASE process outlined on the Student Affairs webpage.

Students participating in Design Competitions supported by Smith Engineering can submit a Request for Excused Absence for Significant Events (REASE) following the process outlined on the [Smith Engineering's Academic Consideration \(Absences\) website](#).

All Requests for Excused Absences for Significant Events must be submitted to the Faculty a minimum of 2 weeks prior to the event start date.

RELIGIOUS ACCOMMODATIONS

Students in need of accommodation for religious observance are asked to complete the [Religious Observance and Academics Activities](#) form within a week of receiving their course syllabus (each term) and submit the form via email to the Engineering Program Advisor, (Accommodations & Considerations), engineering.aac@queensu.ca

- Alternative assignments are considered a “reasonable accommodation” under the Ontario Human Rights Code
- Please refer to the [Queen’s Multi-faith Calendar](#) for all of this year’s faith dates
- Students with questions about their rights and responsibilities regarding religious accommodations can contact chaplain@queensu.ca

OTHER HUMAN-RIGHTS BASED ACCOMMODATIONS

Engineering students who need accommodation on Human Rights grounds are asked to contact, in confidence, the Engineering Program Advisor, (Accommodations & Considerations) engineering.aac@queensu.ca or to call 613-533-6000 x 78013.

ACCOMMODATIONS FOR DISABILITIES

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the [QSAS website](#) to learn more about academic accommodation.

To start the registration process with QSAS, click the “**Access Ventus**” button found on the [Ventus student portal](#).

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students:

<https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students>

For questions or assistance with requesting Academic Consideration or Accommodation, contact the Smith Engineering Program Advisor (Accommodations and Considerations) at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

STUDENT WELLBEING

Success in Engineering includes taking care of your wellbeing. If you're experiencing stress, anxiety, burnout, personal challenges, or you're simply not sure where to turn, [EngWell](#) is here to help.

EngWell provides confidential support through our [Wellness Advisor](#), [Embedded Counsellors](#), [mental health programming](#), and referrals to academic and campus resources. Whether you need short term guidance, help navigating university supports, or someone to talk to, [we encourage you to reach out early](#).

EDUCATIONAL TECHNOLOGY

Educational Technology	Accessibility Compliance Information
onQ (Brightspace Learning Management System by D2L)	https://www.d2l.com/accessibility/standards/
MS-Teams	https://support.microsoft.com/en-us/office/accessibility-support-for-microsoft-teams-d12ee53f-d15f-445e-be8d-f0ba2c5ee68f
Zoom	https://zoom.us/accessibility

If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

TECHNICAL SUPPORT

Some basic comfort levels with basic hardware and software skills are required for this course. If you require technical assistance, please contact [Technical Support](#).